

Current Advances in LLM Reasoning

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Abstract

As Large language models (LLMs) increasingly tackle reasoning-heavy tasks, from mathematics to commonsense to multilingual understanding, researchers face three pressing questions: How well do models reason? How can we make them reason better? and What are the next frontiers in LLM reasoning? This tutorial answers these questions through a unified view of LLM reasoning. This tutorial explores comprehensive evaluation strategies to assess the reasoning abilities of models and discusses two types of methods to improve models' reasoning: advanced inference time methods, such as structured and self-improvement inference methods, and (ii) post-training methods, such as RLHF, DPO, and GRPO that aim to make LLMs think more like humans. The tutorial explores these technical discussions while maintaining a practical outlook through illustrative demos and short guided hands-on exercises. The tutorial is designed for both researchers and practitioners seeking practical insights into LLM reasoning.

1 Introduction

Reasoning has emerged as a central capability in LLMs across mathematical, commonsense, scientific, and other domains. Early prompting-based methods such as Chain-of-Thought (CoT) (Wei et al., 2022) showed that reasoning step-by-step in natural language, rather than relying on end-to-end black-box predictions, can serve as an interpretable and effective reasoning trace. This insight led to a wave of research into (i) advanced inference time methods such as structured inference methods including Tree-of-Thought reasoning (Yao et al., 2023), self-reflection (Shinn et al., 2023), and structured orchestration frameworks such as Fleet of Agents (Klein et al., 2025) and (ii) post-training methods such as RLHF (Ouyang et al., 2022), Direct Preference Optimization (Rafailov et al., 2023), and Group Relative

Policy Optimization (Shao et al., 2024) that aim to make LLMs think more like humans.

This half-day tutorial offers a practical guide to this fast-moving landscape of LLM reasoning. We cover three key questions:

1. *How well can models reason?* We examine how current LLMs reason and how to evaluate their reasoning abilities beyond accuracy. We cover low-resource settings such as low-resource languages, and robustness settings, and discuss evaluation dimensions such as consistency, faithfulness, calibration, and variance-aware scoring.
2. *How do we make models reason better?* We explore algorithmic and systems-level strategies that enhance reasoning capability, from prompting and structure to agentic behavior and post-training reinforcement learning.
3. *How is the path to the the next frontiers in reasoning?* We explore state-of-the-art compute-efficient reasoning strategies, designing inclusive benchmarks and other open research questions.

This tutorial answers these key questions while maintaining a practical outlook through illustrative demos, and short guided hands-on exercises. Each part of the tutorial includes short, guided hands-on exercises using hosted Colab notebooks. Participants will try out evaluation techniques like measuring consistency and calibration, and explore how reasoning varies across languages and random seeds. They will also experiment with different prompting strategies (e.g., iterative, agentic) under latency constraints, and see how reward models affect reasoning confidence. These exercises are designed to be easy to follow and help participants connect theory to practice.

This tutorial differs from prior excellent CL tutorials on LLMs by combining hands-on practical

exercises with a focused exploration of how to improve model reasoning. We refer to prior tutorials for technical LLM foundations, and our emphasis is on actionable methods and insights that enhance reasoning in real-world workflows.

2 Tutorial Structure

The tutorial is organized around three guiding questions that collectively capture the state, practice, and future of reasoning with large language models (LLMs). The three-hour session is split into two 90-minute blocks separated by a 30-minute coffee break. Each part combines concise lectures, illustrative demos, and short guided hands-on exercises. Additionally, each part clearly lists the lead instructors. Please see presenter biographies (Section 11) for their specific areas of expertise and prior experience in instructing international audiences.

Part 1 – How well can models reason? (≈ 60 minutes including 20 minutes hands-on)

Leads: Julia Kreutzer, Akhil Arora, Nearchos Potamitis.

This part establishes a foundation by examining how current LLMs reason and how to evaluate their reasoning abilities beyond accuracy. We cover multilingual, low-resource, and robustness benchmarks (Hou et al., 2025; Cecilia Liu et al., 2024; Chang et al., 2025; Fabbri et al., 2025), and discuss evaluation dimensions such as reasoning language, consistency, faithfulness, calibration, and variance-aware scoring (Liang et al., 2023; Srivastava et al., 2023; ?; Bowman, 2023; Dziri et al., 2021, 2022; Sun et al., 2025; Zhong et al., 2025). Through representative studies, participants see how dataset artifacts, sampling variance, and linguistic diversity impact the interpretation of model “intelligence” (Singh et al., 2025; Romanou et al., 2025; Rystrøm et al., 2025).

Embedded hands-on (20 min): Guided by Nearchos Potamitis, participants use a prepared Colab notebook to assess reasoning variance across languages and seeds, compute calibration error, and visualize consistency metrics—revealing how evaluation methodology shapes perceived capability (Srivastava et al., 2023; ?).

Part 2 – How do we make models reason better? (≈ 90 minutes, 30 minutes hands-on)

Building on the diagnostic insights from Part 1, this part focuses on improving reasoning performance through algorithmic and system-level innovations.

2.1 Inference-time Reasoning Strategies (45 min) Leads: Niket Tandon, Akhil Arora, Nearchos Potamitis. We unify prompt-based, structured, and agentic reasoning under a single lens. Topics include iterative and self-refinement methods (Madaan et al., 2023) and structured prompting via graphs/trees (Wei et al., 2022; Kojima et al., 2022; Yao et al., 2023; Shinn et al., 2023), tool-augmented/agentic reasoning (Yao et al., 2023; Shinn et al., 2023), and dynamic compute allocation (Leviathan et al., 2023; Schuster et al., 2022; Wang et al., 2024). We also examine coordinated and efficient orchestration frameworks that combine planning, reflection, and tool use, discussing practical trade-offs in cost, latency, and quality (Klein et al., 2025; Potamitis et al., 2025).

Embedded hands-on (15 min): Guided by Nearchos Potamitis, participants compare iterative, self-consistent, and agentic prompting strategies under latency budgets, quantifying cost–quality trade-offs (Wei et al., 2022; Yao et al., 2023; Leviathan et al., 2023).

– Coffee Break (30 min) –

2.2 Post-training and RL-based Reasoning (45 min) Leads: Nouha Dziri, Vishrav Chaudhary. This segment focuses on reinforcement-learning and preference-based post-training approaches (RLHF, DPO, reward modeling) that enhance robustness and controllability of reasoning (Ouyang et al., 2022; Rafailov et al., 2023; Ahribian et al., 2025; Karaman et al., 2025). We relate these to inference-time strategies, emphasizing synergy between fine-tuning and test-time reasoning, and include recent alignment variants (Guo and DeepSeek-AI, 2025; Yu et al., 2025).

Embedded hands-on (15 min): Guided by Nearchos Potamitis, participants explore a compact demo showing how reward-model calibration influences reasoning confidence and abstention behavior (Ouyang et al., 2022; Rafailov et al., 2023).

Part 3 – What are the next frontiers in reasoning? (≈ 30 minutes, no hands-on)

3.1 Frontiers for Efficient Reasoning (15 min)

Leads: Akhil Arora, Niket Tandon. We discuss how to make reasoning more sustainable and cost-effective—*green AI friendly*—via adaptive compute reuse, caching, batching, and scalable orchestration (Schuster et al., 2022; Leviathan et al., 2023; Wang et al., 2024; Potamitis et al., 2025; Potamitis and Arora, 2025). Connections to reproducibility and deployment efficiency are highlighted.

3.2 Frontiers for Reasoning in High-Stakes Domains (15min)

Leads: Lars Klein. We explore a transition from structured reasoning to operating over structured data. This segment highlights a spectrum of reasoning typologies: Contrasting abductive reasoning where the solution space is undefined and requires creative hypothesis generation with strictly deductive, closed-form tasks. We discuss a critical paradigm shift necessary for high-stakes deployment: moving from a “reason as hard as possible and then make an educated guess” mentality to one of calibrated abstention. We will explore how to train and evaluate models to recognize their own uncertainty, effectively refusing the temptation to guess when in doubt, thereby prioritizing safety and factual reliability over mere completion.

3.3 Frontiers for Low-Resource and Multilingual Reasoning (15 min)

Leads: Julia Kreutzer, Akhil Arora. We explore advances and open challenges in multilingual and culturally grounded reasoning, including cross-lingual transfer, low-resource adaptation, and inclusive benchmark design (Singh et al., 2025; Romanou et al., 2025; Qi et al., 2025; Chang et al., 2025).

Finally, the tutorial concludes with an integrative discussion of emerging questions—such as improving the stability of reasoning performance and establishing community-standard evaluation protocols—linking research gaps to opportunities for collaboration within the ACL community.

3 Technical Details

Tutorial type and duration. Cutting-edge tutorial in CL/NLP (with hands-on component).¹ Reasoning has become a defining capability of mod-

¹We request a half-day (3-hour) slot at ACL 2026 in San Diego; per the call, this occupies a 3.5-hour block including a coffee break.

ern LLMs, yet current progress remains scattered across evaluation protocols, inference-time strategies, post-training optimization, and multilingual generalization. This tutorial unifies these developments through three guiding questions:

- (1) *How well can models reason?*—covering multilingual, low-resource, and variance-aware evaluation;
- (2) *How do we make models reason better?*—spanning inference-time and post-training approaches, from structured and agentic prompting to RL-based fine-tuning; and
- (3) *What are the next frontiers in reasoning?*—exploring efficiency, sustainability, and low-resource multilingual challenges.

Participants will gain a comprehensive understanding of the state of the art and hands-on experience with frameworks that make reasoning both reliable and efficient.

Target Audience and Prerequisites. The tutorial targets NLP and CL researchers, engineers, and practitioners working on LLM applications, multilingual NLP, evaluation, or reasoning systems. It assumes familiarity with foundational machine learning concepts but does not require specialized background knowledge.

Suggested prerequisites. Participants are expected to have a working knowledge of basic probability and optimization at the level of introductory calculus, familiarity with supervised machine learning and Transformer architectures, and proficiency in Python with some experience using PyTorch, JAX, or Google Colab. A background in linguistics is not required, though an awareness of cross-lingual transfer and typological variation may be helpful.

Breadth and Use of Prior Work. The tutorial balances the presenters’ recent contributions with a broad overview of community progress. Each part integrates representative methods from multiple research groups across reasoning, efficiency, and evaluation. The reading list (Section 5) offers core conceptual primers, while the tutorial outline (Section 2) embeds state-of-the-art works from 2023–2025, ensuring comprehensive and up-to-date coverage. All accompanying code assets will credit the respective frameworks and datasets, and the tutorial foregrounds comparative analyses, open challenges, and community-driven benchmarks rather than individual projects.

4 Diversity, Inclusion, and Accessibility

Content spans multilingual and low-resource settings and discusses cultural grounding and inclusivity in evaluation. The presenter team spans academia/industry, multiple institutions, and countries, and includes junior and senior researchers, and strives to strike a strong gender balance. We will: (i) share all materials publicly; (ii) provide captioned slides and written step-by-step guides; (iii) ensure examples include non-English data; and (iv) coordinate with affinity groups to promote participation.

5 Reading List (Background and Pre-requisites)

To help participants prepare for the tutorial, we provide a short list of background and survey papers covering fundamental concepts in reasoning, inference, and evaluation with large language models (LLMs). These readings are intended as conceptual primers rather than state-of-the-art works, which will be discussed in detail throughout Section 2.

- **General Reasoning in LLMs:** [Bubeck et al. \(2023\)](#) provide early evidence of emergent reasoning in GPT-4, while [Huang and Chang \(2023\)](#) offer a systematic survey of reasoning taxonomies, evaluation setups, and open challenges.
- **Evaluation and Robustness:** [Srivastava et al. \(2023\)](#) present HELM, a holistic benchmark for measuring the capabilities of language models and [Bowman \(2023\)](#) summarize key conceptual and methodological lessons about large language models.
- **Efficiency and Inference Scaling:** Foundational work on efficient inference includes [Schuster et al. \(2022\)](#) on adaptive language modeling and [Leviathan et al. \(2023\)](#) on speculative decoding for fast transformer inference.
- **Multilingual and Cross-lingual Perspectives:** [Kreutzer et al. \(2022\)](#) discuss multilingual quality estimation and evaluation, while [Artetxe et al. \(2020\)](#) provide a grounding in cross-lingual transfer in multilingual encoders.

Note: This list provides conceptual background and surveys. Recent state-of-the-art works are cited within Section 2, where they are introduced and contextualized.

6 Expected Audience Size and Prior Offerings

We expect 250 attendees, given cross-cutting interest in reasoning, systems, and multilinguality within ACL. While we have not offered this exact tutorial before, parts of the material have been presented in invited talks and short courses; the present tutorial unifies and updates content with a strong hands-on component tailored to ACL 2026.

7 Special Requirements (Equipment/Logistics)

Standard projector, microphones, and reliable Wi-Fi; power strips for front-row tables used during the hands-on; permission to distribute QR codes/links to Colab repositories. No other special equipment required.

8 Open Access of Materials

We will release slides, annotated notebooks, sample datasets, and an instructor’s guide under permissive licenses. We will submit slides and bibliography to the ACL Anthology, and we consent to recording and captioning; all materials will be accessible one month before the tutorial date.

9 Ethics Statement

Reasoning methods influence decision-making and can amplify cultural/linguistic biases. We will cover risk factors (spurious heuristics, evaluation leakage, demographically-skewed errors), emphasize transparent reporting (metrics, datasets, statistical tests), and demonstrate safe deployment patterns (abstention, fallback, and verification). Hands-on materials will avoid sensitive personal data and include multilingual examples with documented licenses.

10 Venue and Limitations

We request ACL 2026 (San Diego) as the preferred venue due to (i) alignment with instructors’ teaching calendars and (ii) visa stability for US-based co-presenters. The tutorial is fully compatible with hybrid participation.

Additional details

Special conference themes: The official conference themes are yet to be announced. We will align the tutorial if themes overlap with our core topics of multilinguality, evaluation, or sustainable and efficient AI systems. Given ACL’s emphasis on responsible and globally inclusive NLP, our focus on *multilingual reasoning* and *variance-aware evaluation* fits naturally within anticipated themes such as fairness, inclusivity, and efficient modeling.

Pedagogical package for classrooms: We will release two self-contained classroom-ready exercises aligned with the tutorial’s three guiding questions: (1) evaluating reasoning variance across prompts and languages (Part 1), and (2) exploring inference-time efficiency trade-offs using structured and agentic prompting (Part 2). Each exercise will include clear instructions, rubrics, and discussion questions, suitable for graduate-level NLP or AI courses.

11 Tutorial Presenters

Akhil Arora (CLAN@AU) is a professor of computer science at Aarhus University (AU), where he leads the [CLAN for AI Research on Language and Networks](#) (or “CLAN” for short). He is a fellow of the Copenhagen Center for Social Data Science, an affiliate of the Pioneer Centre for AI, and a formal collaborator of the Wikimedia Foundation. Akhil’s research lies broadly in human-centered AI with a focus on improving human knowledge-seeking, bridging knowledge gaps, and promoting knowledge equity on the Web. His work blends methods from NLP, AI, Graph ML, and computational social science. For this tutorial, Akhil co-leads Part 1: evaluation and benchmarking, Part 2: inference-time reasoning strategies, and Part 3: next frontiers in reasoning. His recent research includes frameworks for cost-efficient and reliable LLM inference (e.g., *CacheSaver* (Potamitis et al., 2025)) and the SOTA for structured reasoning, i.e., *Fleet of Agents* (Klein et al., 2025). He has extensive experience teaching at international graduate summer schools and has organized tutorials and workshops at top-tier venues such as ACL, SIGMOD, EDBT, and DSA. (Email: akhil.arora@cs.au.dk)

Vishrav Chaudhary (Meta Superintelligence) is a Research Scientist working on post-training research and reasoning capabilities in large and small LMs. His research spans efficient multilingual modeling, reinforcement learning from human feedback, and cross-lingual reasoning. He leads the tutorial’s segment on post-training and RL-based reasoning. Vishrav has over a decade of experience leading major multilingual modeling efforts at Meta, Microsoft Turing, and FAIR, including contributions to the LLaMA-4, Phi3, and Phi4 model families. He has delivered invited tutorials and workshops at ACL and NAACL (including the ACL 2023 tutorial on Multilingual LLMs) and served as Area Chair for multiple ACL conferences. (Email: vishrav@meta.com)

Julia Kreutzer (Cohere Labs) is a Senior Research Scientist focusing on multilingual reasoning and evaluation for LLMs. She has previously worked at Google Translate and obtained her PhD from the University of Heidelberg. Julia has led research at the intersection of multilingual evaluation, post-training, and inclusivity in language technologies. In this tutorial, she co-leads Part 1:

evaluation and benchmarking) and Part 3: frontiers for low-resource reasoning. Julia has co-organized COLM and multiple ACL workshops, and she has extensive experience teaching and mentoring international researchers through open-science initiatives and global academic programs. (Email: juliakreutzer@cohere.com)

Nearchos Potamitis (CLAN@AU) is a PhD student in Computer Science at Aarhus University (AU), advised by Akhil Arora. His research explores reasoning, efficiency, and evaluation in LMs, focusing on practical systems for robust and affordable inference. In this tutorial, he coordinates all hands-on components and contributes to Parts 1 and 2. Nearchos has prior experience developing educational materials and interactive demos for graduate NLP courses at AU and for international summer schools. His work has appeared in top ML and NLP venues, including ICML and EMNLP. (Email: nearchos.potamitis@cs.au.dk)

Lars Klein (EPFL, LiGHT) is Academic Director at the Laboratory for Intelligent Global Health and Humanitarian Response Technologies at EPFL. He contributes to Part 3: reasoning in high-stakes domains. Before joining the team at LiGHT, Lars has obtained a PhD at EPFL, working on innovative human-computer-interfaces as well as foundational research on structured reasoning and AI. His work has appeared in The Web Conference, ICML, EMNLP and includes state-of-the-art advances in structured reasoning. (Email: lars.klein@epfl.ch)

Nouha Dziri (AI2) is a Research Scientist at the Allen Institute for AI (AI2) working with Yejin Choi. Her research spans post-training, reasoning robustness, and LLM interpretability. She co-leads Part 2: post-training and RL-based reasoning. Nouha has co-led the OLMo safety and post-training teams and previously worked at Google Research, Microsoft Research, and Mila. She has organized workshops at ICLR, NeurIPS, and ACL, and regularly teaches short courses and invited sessions on reasoning and safety in LLMs to international research audiences. (Email: nouhad@allenai.org)

Niket Tandon (Microsoft Research) is a Principal Research Scientist focusing on feedback-guided and self-reflective reasoning in LLMs. He co-leads Part 2: inference-time reasoning strategies and contributes to Part 3: efficient reason-

ing frontiers. Niket previously led reasoning research at AI2's Aristo team and earned his PhD from MPI for Informatics under Gerhard Weikum. His work has appeared in ACL, NeurIPS, and EMNLP, and includes the creation of large-scale commonsense resources like WebChild. Niket has organized and taught multiple international tutorials and workshops, including those at NeurIPS and EMNLP, and leads outreach initiatives for mentoring undergraduates from underrepresented backgrounds through PQRS Research. (Email: niket.tandon@microsoft.com)

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